

Article

Deploying Efficient LLM Agents on Maritime Autonomous Surface Ships: Fine-Tuning, RAG, and Function Calling in a Mid-Size Model

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Abstract

Deploying Large Language Models (LLMs) on Maritime Autonomous Surface Ships (MASS) entails a critical trade-off between reasoning depth, inference latency, and hardware constraints. To fill the existing gap, we introduce **MARTIAN** (Maritime Agent for Real-time Tactical Inference And Navigation), a 14B-parameter decision support agent engineered for edge deployment on standard vessel hardware (e.g., the NVIDIA Jetson AGX Orin). Central to our approach is the Cognitive Core architecture, which utilizes a verified dataset of 21,800 Chain-of-Thought (CoT) instruction–response pairs to align general linguistic capabilities with maritime procedural logic. Empirical evaluations demonstrate that MARTIAN achieves an overall accuracy of **73.23%** (SFT only) and **81.16%** (SFT + RAG) on the Bilingual Maritime Multiple-Choice Questionnaire (BM-MCQ), a standardized assessment dataset constructed based on Officer of the Watch (OOW) competencies. Notably, the SFT-only configuration attains **78.53%** on pure-logic-intensive COLREG tasks—surpassing the 72B-parameter Qwen-2.5 foundation model in this domain—while maintaining a real-time inference latency of **22.4 ms/token**. Crucially, our ablation studies support a nuanced *Interference Hypothesis*: while RAG significantly enhances factual recall in knowledge-intensive domains (boosting total accuracy from 73.23% to 81.16%), it concurrently introduces semantic noise that degrades performance in pure logic reasoning tasks (e.g., COLREG maneuvering accuracy decreases from 78.53% to 77.36%). On the basis of this finding, we identify and empirically motivate a *decoupled cognitive design principle* that separates procedural reflexes (via SFT) from declarative knowledge (via RAG). While the full implementation of an adaptive routing mechanism is deferred to future work, the ablation results presented herein offer a validated, cost-effective reference architecture for deploying transparent and regulation-compliant AI on resource-constrained merchant vessels.



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1. Introduction

The transition towards Maritime Autonomous Surface Ships (MASS) is accelerating, driven by the imperative to enhance safety and efficiency in global logistics. Despite significant advances in environmental perception utilizing radar, the Automatic Identification System (AIS), and visual sensors [1,2], maritime operations remain heavily reliant

on human supervision. Statistical investigations reveal that most maritime accidents are attributable to erroneous human decisions, often stemming from fatigue, negligence, or the misinterpretation of complex regulations in dynamic multi-vessel encounters [3–5]. Consequently, bridging the semantic gap between low-level sensor perception and high-level cognitive reasoning remains a paramount research priority in the domain of intelligent transportation [6].

Traditional approaches to autonomous decision-making have exhibited distinct limitations. Rule-based systems, while deterministic, often struggle to resolve the ambiguity inherent in real-world scenarios or to interpret the nuanced provisions of the *International Regulations for Preventing Collisions at Sea* (COLREGs) [7,8]. Conversely, data-driven methods based on Deep Reinforcement Learning (DRL) or Artificial Potential Fields (APF) [9–11], though adaptable, frequently function as black boxes. This lack of transparency and explainability poses a significant barrier to their certification for safety-critical maritime operations, where stakeholders require explicit rationales for maneuvering decisions [12].

The advent of the Transformer architecture [13] and Large Language Models (LLMs) has introduced a promising paradigm for embodied reasoning. Recent studies have demonstrated the potential of LLMs in autonomous driving [14,15] and robotics [16], leveraging their emergent capabilities in common-sense reasoning and zero-shot generalization [17]. In the maritime domain, pioneering efforts such as the development of NavGPT [18] have begun to explore the use of LLMs to interpret navigational scenarios. However, deploying these capabilities aboard vessels presents substantial engineering challenges. First, state-of-the-art performance is typically associated with massive cloud-based models (e.g., GPT-4o [19]), which depend on satellite connectivity that is often unstable and bandwidth-constrained at sea. Second, smaller local models often lack the specific domain knowledge required for compliant maneuvering; while techniques such as Low-Rank Adaptation (LoRA) [20] and quantization [21,22] offer pathways for edge deployment, the optimal architecture for balancing inference latency, factual accuracy, and reasoning depth remains underexplored.

To address these constraints, this paper introduces **MARTIAN** (**M**aritime **A**gent for **R**ead-time **T**actical **I**nterference **A**nd **N**avigation), an edge-optimized decision support system designed for standard vessel hardware (see Figure 1). Rather than relying on a large monolithic model, we investigate a composite cognitive architecture that integrates three specialized mechanisms: (1) **Supervised Fine-Tuning (SFT)** [23] to provide the system with procedural maneuvering logic; (2) **Retrieval-Augmented Generation (RAG)** [24,25] to ground decisions in regulatory texts; and (3) **Function Calling** [26] for deterministic sensor integration.

Crucially, we analyze the interaction between these mechanisms beyond simple accumulation. Recent findings in reasoning-optimized models, such as DeepSeek-R1 [27], suggest that external knowledge retrieval might interfere with internalized reasoning chains. Our experiments corroborate this *Interference Hypothesis* in the maritime context: while RAG significantly enhances factual recall for static regulations, it may introduce semantic noise that degrades performance in dynamic, logic-heavy maneuvering tasks. It is important to note that MARTIAN is designed as a *decision support agent* rather than a fully autonomous controller. It excels at parsing situational data and proposing compliant maneuvers (reasoning and acting), while complex multi-step strategic planning remains under human supervision. The main contributions of this work can be summarized as follows:

1. **Feasibility Verification:** We confirm that a 14B-parameter model, optimized via INT4 quantization and FlashAttention [28], can achieve 81.16% overall accuracy (SFT + RAG) and 78.53% on logic-intensive COLREG tasks (SFT only) while maintaining a real-time

decoding latency of 22.4 ms/token on embedded hardware (NVIDIA Jetson AGX Orin, 64 GB).

2. **Comparative Analysis:** We conduct a rigorous ablation study—including domain-specific fine-tuned baselines to ensure comparison fairness—that reveals the trade-offs between SFT and RAG. The results demonstrate that SFT is superior for establishing *procedural reflexes* in collision avoidance, whereas RAG is essential for *declarative knowledge* queries. We quantify this interference effect through reasoning perplexity and logical breakage metrics.
3. **Dataset Release:** We present a verified dataset of 21,800 CoT instruction–response pairs [29], strictly aligned with the Standards of Training, Certification and Watchkeeping (STCW) competencies, to facilitate further research in trustworthy maritime AI.

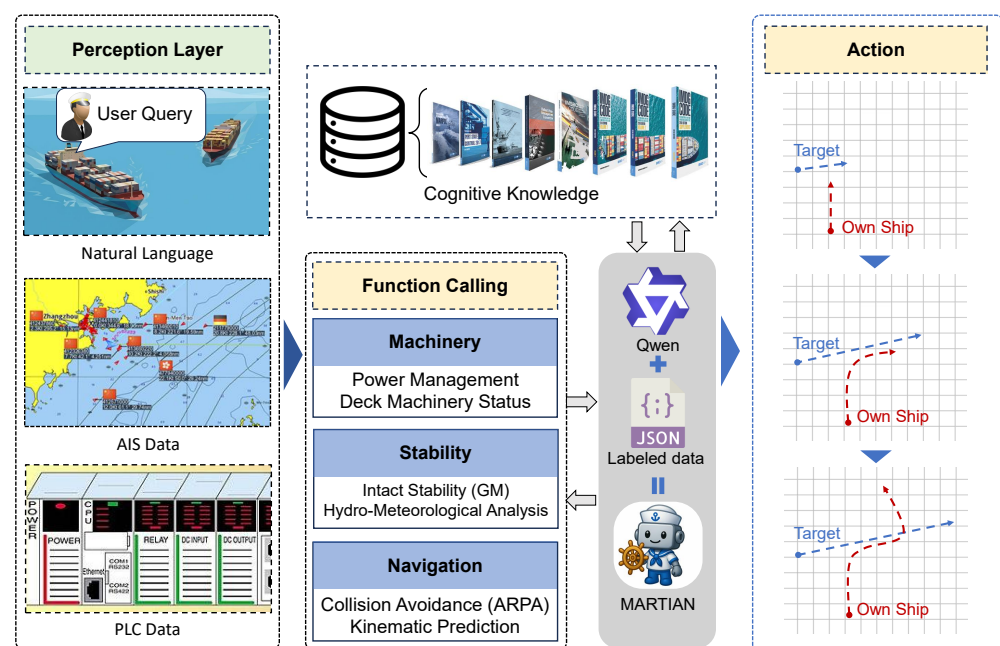


Figure 1. The architectural framework of MARTIAN. (Left) The edge-deployed perception interface aggregates multimodal inputs, including natural language queries, AIS data streams, and Programmable Logic Controller (PLC) telemetry. (Center) The cognitive core integrates three specialized mechanisms: (1) a function calling module for real-time vessel monitoring (navigation, stability, machinery); (2) a cognitive knowledge base (RAG) for regulatory grounding; and (3) an SFT-optimized reasoning engine (Qwen-2.5) for decision synthesis. (Right) An illustrative workflow of a COLREG crossing scenario. The agent detects a target on the port bow, retrieves Rule 15/16 via RAG, and executes a compliant course alteration through Chain-of-Thought reasoning.

2. Methodology

The MARTIAN architecture is designed as a centralized cognitive core for MASS. As depicted in Figure 1, it addresses the dual challenges of edge-computing constraints and the need for explainable reasoning by integrating three synergistic modules: a CoT-optimized reasoning engine, a retrieval-augmented generation (RAG) system for regulatory grounding, and a function calling interface for real-time sensory perception.

2.1. Construction of the Maritime Chain-of-Thought Dataset

A primary barrier to maritime AI is the lack of public, instruction-based datasets. To bridge this gap, we constructed a specialized dataset of 21,800 instruction–response pairs aligned with STCW competencies. Unlike generic QA datasets, our focus was on synthesizing CoT reasoning traces that emulate the decision-making process of an expert Officer of the Watch (OOW).

As illustrated in Figure 2, the construction process followed a three-stage knowledge distillation pipeline. We first aggregated heterogeneous data sources, ranging from the text of COLREGs to anonymized maneuvering logs. A teacher model (GPT-4o) was then employed to distill this declarative knowledge into procedural reasoning traces, ensuring the explicit generation of rationale before commands. Finally, to mitigate hallucination, all synthetic traces were processed through a rigorous hybrid validation pipeline, as follows. **Stage 1: Automated Filtering.** Each generated CoT trace was screened by a suite of regex-based validators targeting (i) structural completeness (presence of <Analysis>, <Action>, and <Rationale> tags), (ii) numerical consistency (e.g., verifying that cited rule numbers fall within valid COLREG/SOLAS ranges), and (iii) logical coherence (e.g., ensuring the recommended action does not contradict the cited regulation). Of the 28,500 initially generated pairs, 4120 (14.5%) were rejected at this stage owing to missing reasoning steps or referencing non-existent regulations. **Stage 2: Semantic Deduplication.** Near-duplicate pairs were identified using MinHash-based locality-sensitive hashing (Jaccard threshold > 0.85). An additional 2580 redundant traces were removed, yielding 21,800 unique instruction pairs. **Stage 3: Expert Sampling and Annotation.** A stratified random sample of 2000 traces (approximately 9.2% of the final dataset) was independently reviewed by two certified OOW-qualified maritime officers. Each trace was assessed on a three-point scale (Correct/Partially Correct/Incorrect) along three dimensions: factual accuracy, procedural validity, and reasoning coherence. The inter-annotator agreement, measured by Cohen's κ , was 0.81, indicating substantial agreement. Discrepancies were resolved through adjudication by a senior examiner. The sampled accuracy rate was 94.3%, with the dominant mode of failure being incomplete (but not incorrect) reasoning chains. **Teacher Model Prompting.** The GPT-4o teacher model was prompted with a structured template containing: (i) the source regulation text or scenario description; (ii) a role specification ("You are a senior Officer of the Watch..."); and (iii) explicit instructions to generate step-by-step reasoning *before* any action output, formatted as follows: [Situation] $\rightarrow$ [Applicable Rules] $\rightarrow$ [Analysis] $\rightarrow$ [Action]. Temperature was set to 0.3 to balance diversity with factual fidelity.

Representative samples covering navigation, safety, and operations are instantiated in the case study shown in Figure 2b, demonstrating the model's capability to resolve dynamic collision risks.

2.2. Cognitive Specialization via Supervised Fine-Tuning

To deploy this reasoning capability on resource-constrained vessel hardware, we adopted the **Qwen-2.5-14B-Instruct** as our base model; while large foundation models offer broad knowledge, they lack specific maritime heuristics.

We implemented SFT using LoRA. Specifically, for a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, the update is constrained by a low-rank decomposition $W_0 + \Delta W = W_0 + BA$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are trainable parameters with rank $r \ll \min(d, k)$. The forward pass is formalized as

$$h = W_0 x + \frac{\alpha}{r} B A x, \quad (1)$$

where α is a scaling factor constant. In our empirical setup for MARTIAN, we configured the low-rank dimension to $r = 16$ and the scaling factor to $\alpha = 32$. The adapters were injected into the attention mechanisms and feed-forward projections of the base model. Instead of updating all 14 billion parameters, this strategy allows the model to internalize the procedural logic from our dataset—effectively adapting its probabilistic distribution to maritime contexts—while maintaining a memory footprint small enough for consumer-grade GPUs.

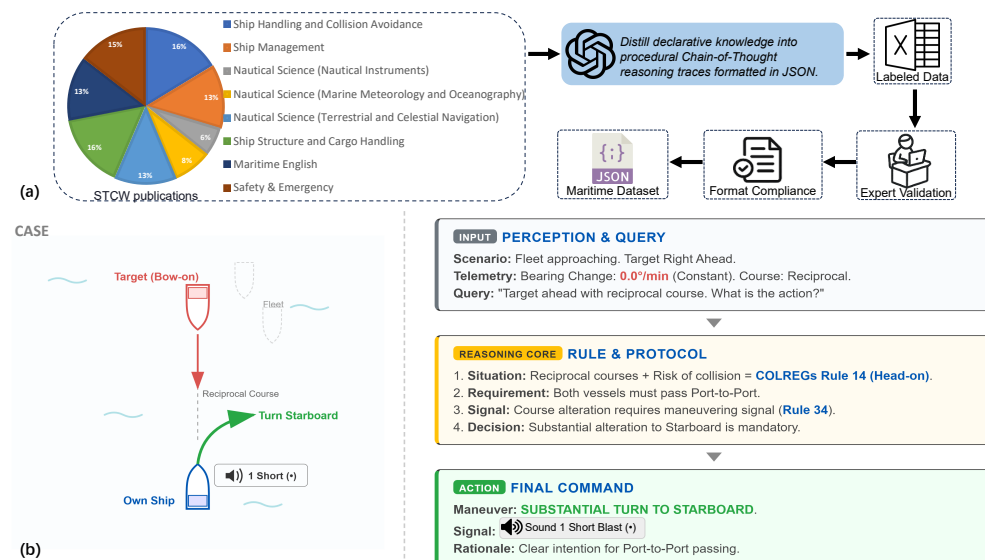


Figure 2. Maritime CoT dataset construction and qualitative analysis. (a) The three-stage automated synthesis pipeline transforms static STCW declarative knowledge into procedural reasoning traces. (b) A representative head-on scenario (Rule 14) generated by the pipeline. The visualization demonstrates the model’s capacity to explicitly coordinate kinematic maneuvers (Substantial Turn) with required sound signals (Rule 34) through a structured Chain-of-Thought process.

2.3. Operational Grounding via Function Calling

Standard LLMs suffer from a perceptual limitation—they are disconnected from the vessel’s physical state. We address this through a robust function calling module that serves as the agent’s perception interface. Unlike generic API triggers, this module is designed as a library of deterministic engineering primitives, strictly grounding the agent’s reasoning in nautical mathematics.

The interaction logic follows the **LLM tool protocol** defined in Algorithm 1. The agent autonomously parses natural language queries and selects the appropriate mathematical tool to synthesize a situational report.

To ensure comprehensive coverage, we implemented eight core primitives categorized into the **navigation, hydrodynamics, and machinery** domains. The specific mathematical basis for each tool—such as the Barrass formula for squat calculation or the International Maritime Organization (IMO) criteria for stability—is detailed in Table 1.

To ensure reliability, the module includes a rigorous **data pre-processing pipeline**, utilizing the interquartile range (IQR) method to filter statistical outliers from raw sensor streams and applying linear interpolation to align asynchronous timestamps.

Table 1. Library of deterministic maritime tools implemented in MARTIAN.

Category	Function Primitive	Mathematical/Logical Basis	Input Source
Navigation and Risk	calc_cpa_tcpa	Vector calculation: $\mathbf{p}_{rel} \cdot \mathbf{v}_{rel}$ to determine CPA/TCPA.	Radar/AIS
	calc_squat	Barrass formula: $S_{max} = C_b \cdot V_k^2 / 100$ for UKC assessment.	Echo sounder
	pred_crash_stop	Kinetic model using displacement Δ and resistance k .	Speed log
Stability and Maneuvering	calc_stability_gm	IMO formula: $GM = (f \cdot B / T_\phi)^2$ derived from roll period.	Gyro/IMU
	pred_turning	Nomoto Model: est. radius $R \approx V / \omega$ based on rudder angle.	Steering gear
Operations and Env.	calc_energy	Time-domain integration: $E = \int P(t) dt$.	Power mgmt.
	calc_wind_load	OCIMF standard: lateral force F_{wind} calculation.	Anemometer
	query_specs	Exact-match lookup against vessel technical manuals.	Static DB

Note: CPA = closest point of approach; TCPA = time to CPA; UKC = under-keel clearance; GM = metacentric height.

Algorithm 1 The MARTIAN Operational Grounding Protocol.**Require:** Query Q , Toolset $\mathcal{T}$, Sensor Context $\mathcal{C}_{\text{sensor}}$, Knowledge Base $\mathcal{K}_{\text{RAG}}$ **Ensure:** Grounded Response R **// Phase 1: Intent Parsing & Tool Selection**1: $\mathcal{A} \leftarrow \text{LLM}_{\text{Router}}(Q, \mathcal{C}_{\text{sensor}}, \mathcal{T})$ 2: $\mathcal{C}_{\text{know}} \leftarrow \text{Retrieve}(Q, \mathcal{K}_{\text{RAG}})$ 3: **if** $\mathcal{A} = \emptyset$ **then**4: **return** $\text{LLM}_{\text{Reason}}(Q, \mathcal{C}_{\text{know}})$ 5: **end if****// Phase 2: Deterministic Execution & Validation**6: Initialize observation set $\mathcal{O} \leftarrow \emptyset$ 7: **for all** $(t, \text{args}) \in \mathcal{A}$ **do**8: **try**9: $\text{raw_val} \leftarrow \text{Execute}(t, \text{args})$ 10: **if** $\text{Validate}_{\text{IQR}}(\text{raw_val}, \mathcal{C}_{\text{sensor}})$ **then**11: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\text{Format}(t, \text{raw_val})\}$ 12: **else**13: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\text{Format}(t, \text{Warning: Data Outlier})\}$ 14: **end if**15: **catch** $\text{ExecutionError } e$ 16: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\text{Format}(t, \text{Error: } +e)\}$ 17: **end try**18: **end for****// Phase 3: Contextual Synthesis**19: $P_{\text{final}} \leftarrow \text{Prompt}(Q, \mathcal{C}_{\text{sensor}}, \mathcal{C}_{\text{know}}, \mathcal{O})$ 20: $R \leftarrow \text{LLM}_{\text{Generate}}(P_{\text{final}})$ 21: **return** R **2.4. Knowledge Injection via Retrieval-Augmented Generation (RAG)**

While SFT provides procedural reasoning patterns, ensuring strict adherence to static regulations (e.g., exact COLREG phrasing) requires external memory. Our RAG module utilizes a hybrid search strategy over a vectorized knowledge base of 15 maritime domains.

Crucially, to maintain offline availability on edge devices, documents are segmented and encoded using local high-dimensional embeddings (specifically, **BGE-M3**). At inference time, relevancy is scored using a cosine similarity metric:

$$S(q, d) = \frac{\mathbf{v}_q \cdot \mathbf{v}_d}{\|\mathbf{v}_q\| \|\mathbf{v}_d\|} \quad (2)$$

where $\mathbf{v}_q$ and $\mathbf{v}_d$ represent the vector embeddings of the query and document chunk, respectively. The system retrieves the top K chunks ($K = 3$) to act as a factual anchor, minimizing hallucination in safety-critical answers.

3. Experiments

To thoroughly validate the MARTIAN architecture, we designed a comprehensive evaluation framework covering two critical dimensions: (1) cognitive reasoning capability as measured by a standardized maritime benchmark and (2) operational interaction proficiency via function calling on real vessel logs. Additionally, we incorporated a hardware-in-the-loop (HIL) replay evaluation on embedded vessel hardware and a scenario-based comparison with classical non-LLM baselines to establish deployment feasibility and architectural necessity.

3.1. Experimental Setup

To ensure the proposed MARTIAN architecture is both operationally feasible and cognitively robust, we established an evaluation environment focusing on hardware efficiency, dataset diversity, and comparative benchmarking.

Hardware and Software Configuration. The fine-tuning phase was conducted on a multi-GPU workstation (4× RTX-4090) running Ubuntu 22.04 and CUDA 12.1. For edge deployment evaluation, MARTIAN was deployed on an NVIDIA Jetson AGX Orin (64 GB) module configured with JetPack SDK 6.0, CUDA 12.2, and PyTorch 2.3.0. The model weights were quantized to INT4 precision for deployment, and FlashAttention was enabled via the TensorRT-LLM back end to optimize memory access during attention computation.

Training and Inference Settings. During Supervised Fine-Tuning, the model was trained using AdamW with a cosine learning rate schedule (initial rate 2×10^{-5} , 3% warmup), global batch size 16, and maximum sequence length 2048, stabilized by bfloat16 mixed precision and gradient clipping (norm 1.0). **LoRA Configuration.** We applied LoRA to transformer attention and MLP projections with rank $r = 16$, scaling factor $\alpha = 32$, and dropout 0.05. For inference and evaluation, decoding parameters were fixed to temperature 0.7, top p 0.9, and a maximum generation length of 512 new tokens.

Evaluation Tasks and Datasets. We assessed the agent by using **BM-MCQ** questions (750 multiple-choice items) to assess cognitive/regulatory reasoning in four domains: *operational seamanship*, *collision avoidance (COLREGs)*, *navigational systems*, and *vessel management*. **To ensure evaluation integrity, strict data partitioning was enforced:** the 750 BM-MCQ items were drawn from official STCW competency examination banks that were *excluded* from the 21,800 CoT training pairs. No question stem, answer option, or paraphrased variant from BM-MCQ appeared in the fine-tuning corpus. We further verified the absence of overlap by computing ROUGE-L similarity between every test question and all training instructions, confirming that no pair exceeded a threshold of 0.7. To anchor the BM-MCQ against an established external reference, we additionally evaluated MARTIAN and key baselines on a 200-item subset sampled from the MMLU [30] professional knowledge domains (specifically *professional medicine*, *professional law*, and *electrical engineering*), which share the multiple-choice format and domain-specific reasoning requirements. This external benchmark provides a calibration point to contextualize the BM-MCQ difficulty level: MARTIAN achieved 62.5% on MMLU-Pro (vs. 68.1% for Qwen-2.5-14B and 74.3% for Qwen-2.5-72B), confirming that BM-MCQ discriminates at a comparable professional difficulty tier.

Baselines and Comparison Models. For the BM-MCQ, we benchmarked MARTIAN against foundation models including DeepSeek-R1, DeepSeek-V3, Llama-3.1-70B, and the Qwen-2.5 series, plus cloud models (Qwen-Max/Qwen-Plus) as an upper bound. **To ensure a fair comparison, we additionally fine-tuned Qwen-2.5-7B on the same 21,800-pair CoT training set using identical LoRA hyperparameters** (a condition denoted as Qwen-2.5-7B-SFT), thereby isolating the contribution of model capacity from that of domain adaptation. For the encounter scenarios, we implemented a classical non-LLM baseline: a deterministic Rule-Based Expert System (RBES), evaluated on the semantic resolution of the set of encounter scenarios (Figure 3).

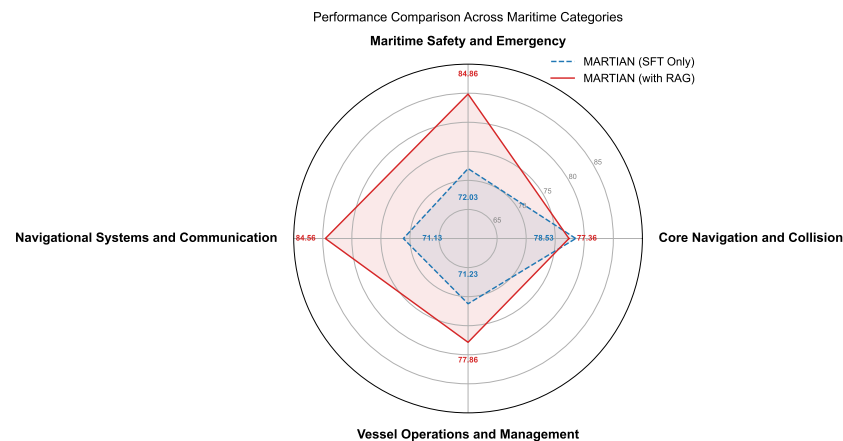


Figure 3. Impact of RAG on domain competencies (ablation study). RAG boosts knowledge-intensive domains (safety and navigational systems) while introducing semantic interference in logic-strict tasks (core navigation), motivating a decoupled design for procedural vs. declarative components.

3.2. Experiment 1: Maritime Knowledge Assessment

To quantitatively evaluate the reasoning capabilities of MARTIAN, we conducted a comparative analysis using BM-MCQ. The results are reported in Table 2.

Table 2. Comparative performance (% accuracy) of MARTIAN against baseline models across four maritime domains. **Bold** indicates the best performance among specialized edge models.

Model	RAG	Overall	Core Nav	Safety	Nav Sys	Vessel Ops
<i>Large-Scale Foundation Models (>70B/Cloud)</i>						
Qwen-Max	✓	82.86	78.36	86.86	86.56	79.66
Qwen-2.5-72B	✓	82.01	77.81	85.91	85.61	78.71
Qwen-Plus	✓	79.75	75.65	83.15	82.55	77.65
DeepSeek-V3	✓	76.63	73.13	80.13	79.13	74.13
Llama-3.1-70B	✓	56.52	53.42	59.32	59.02	54.32
<i>Reasoning-Specialized Models</i>						
DeepSeek-R1	✓	66.01	62.03	69.07	68.57	64.51
DeepSeek-R1	✗	69.26	75.06	67.76	66.84	67.36
<i>Edge-Compatible Baselines (<35B)</i>						
Qwen-2.5-32B	✓	77.76	73.86	81.16	80.76	75.26
Qwen-Turbo	✓	71.81	68.11	74.91	74.21	70.01
Qwen-2.5-14B	✓	71.81	68.31	74.51	74.71	69.70
Qwen-2.5-7B	✓	69.69	66.49	72.59	72.79	66.90
Qwen-2.5-7B-SFT	✗	70.84	72.16	69.42	69.11	72.68
Qwen-2.5-7B-SFT	✓	73.32	71.93	78.41	78.02	72.92
<i>Proposed Method (14B)</i>						
MARTIAN (Ours)	✗	73.23	78.53	72.03	71.13	71.23
MARTIAN (Ours)	✓	81.16	77.36	84.86	84.56	77.86

3.3. Experiment 2: Operational Interaction Proficiency

To demonstrate the agent's capability in handling real vessel logs, we analyze three representative scenarios. These cases highlight strict numerical accuracy in data retrieval, temporal alignment across subsystems, and deterministic boundaries in the event of missing data.

Case 1: Precision Telemetry Retrieval. Query: "At 04:29 on 8 September 2024, what was the RPM of Diesel Generator No.1?" The agent correctly maps the entity to dg_1, invokes get_telemetry, and extracts the exact value without generating unsupported data.

Case 2: Temporal Event Correlation. Query: *Identify key actions that occurred simultaneously on 24 August 2024.* The agent invokes `get_event_log` over the 24-h window and aligns timestamps across Deck Machinery and Positioning subsystems, reporting concurrent events (e.g., crane power-off with DP OFF_DP).

Case 3: Handling of Missing Data (Boundary Analysis). Query: *Average energy consumption of the Side Thruster during DP operations.* The tool retrieves valid power readings but detects that data are missing for some time intervals. The agent requests verification of the sampling interval rather than outputting an unverified energy integration.

3.4. Experiment 3: Hardware-in-the-Loop (HIL) Replay on Embedded Vessel Hardware

Motivation. To bridge the validation gap between textual benchmarks and physical deployments, we evaluate MARTIAN in a hardware-in-the-loop (HIL) setting. The full agent runs on the Jetson AGX Orin while consuming time-stamped vessel logs collected during real operations, thereby verifying execution constraints on the target embedded hardware.

HIL Protocol. We replayed raw telemetry/event logs according to the original `csvTime` stamps. During replay, the agent processed the 100-query equipment status set under strict edge-computing constraints. To ensure statistical reliability, each query was executed over three independent runs. We measured the following: (1) token generation latency (ms/token) during the decoding phase; (2) end-to-end response time t_{e2e} (p_{50} and p_{95}), defined strictly as the total wall-clock time from query input to the final generated token, inclusive of prompt pre-filling, RAG retrieval overhead, tool execution latency, and an average generative length of 120 tokens; (3) tool-call success rate, (4) JSON schema validity rate, and (5) peak GPU memory.

Reporting. Table 3 summarizes the measured responsiveness and stability under streaming conditions. This HIL evaluation verifies that the complete MARTIAN stack sustains real-time constraints on embedded vessel hardware while remaining grounded in deterministic tool outputs.

Table 3. HIL replay performance on NVIDIA Jetson AGX Orin (64 GB).

Metric	Measured	Unit
Token latency (INT4)	22.4	ms/token
End-to-end response time (p_{50})	3.8	s
End-to-end response time (p_{95})	6.5	s
Tool-call success rate	98.0	%
JSON schema validity	99.0	%
Peak GPU memory	6.5	GB

3.5. Experiment 4: Comparison with Classical Non-LLM Methods

Classical maritime algorithms rely heavily on hard-coded decision trees; while effective for continuous spatial path planning, they exhibit structural limitations when processing semantic ambiguity and multi-rule conflicts. To contextualize MARTIAN's cognitive architecture against non-LLM methods, we conducted a comparative analysis focused strictly on the discrete decision-making layer.

Experimental Design. We constructed a deterministic Rule-Based Expert System (RBES) baseline, encoding strict COLREG quantitative thresholds (CPA/TCPA limits; sector-based encounter classification; and priority rules for crossing, overtaking, and head-on situations). Both the RBES and MARTIAN were evaluated on a subset of 150 complex encounter scenarios extracted from the BM-MCQ, requiring the resolution of rule hierarchy conflicts.

Important Caveat. We acknowledge that this comparison assesses a fundamentally different capability than traditional collision avoidance benchmarks. The RBES was designed for *parameterized numerical inputs* (bearing, range, and speed vectors), whereas the evaluation scenarios are presented in *natural language*. Consequently, the performance of the RBES reflects its inability to parse semantic descriptions rather than a deficiency in its underlying decision logic. To partially address this, we manually extracted structured parameters and fed them to the RBES for a control subset of 30 scenarios where unambiguous numerical data was available. On this structured subset, the RBES achieved 76.7% accuracy, confirming that its poor overall score stems from the NLU gap rather than flawed maritime logic. This comparison therefore specifically demonstrates the necessity of an NLU-capable reasoning layer *upstream* within deterministic planners.

Evaluation Metrics. Performance was quantified by **Semantic Resolution Accuracy**, defined as the success rate in selecting the correct maneuver decision when input conditions featured textual ambiguity or nested conditional constraints.

Results and Analysis. Table 4 demonstrates that the classical RBES baseline underperformed significantly, achieving only 42.7% accuracy. Failure analysis indicated that rigid decision trees systematically deadlocked when descriptive texts lacked precise numerical parameters. Conversely, MARTIAN successfully resolved these scenarios, confirming that neural architectures are necessary for parsing semantic complexity prior to downstream kinematic execution.

Table 4. Comparative decision accuracy between MARTIAN and a classical Rule-Based Expert System.

Decision Architecture	Semantic Resolution Accuracy (%)
Rule-Based Expert System (RBES)	42.7
MARTIAN (SFT only)	78.0
MARTIAN (with RAG)	77.3

3.6. Experiment 5: Operational Safety Metrics Under Extreme Noise

To address robustness and systematic failure analysis in adverse scenarios, we designed a stress test on the function calling pipeline, evaluating the agent’s ability to discriminate between valid telemetry and corrupted sensor streams.

Noise Injection Protocol. We established a baseline control group of 100 queries executed on unedited, nominal telemetry logs. Subsequently, we applied two deterministic perturbations to a separate set of 100 queries: (1) **Temporal Dropout**: deleting 20% of timestamps within queried windows (50 queries), simulating network packet loss. (2) **Physical Outliers**: overwriting nominal continuous values with physically impossible extremes, such as RPM > 10,000 (50 queries), simulating acute sensor calibration failures.

Safety Metrics. We measured three critical safety indicators: (1) **Deterministic Fall-back Rate (DFR)**: the fraction of corrupted queries where the agent successfully triggered the IQR validation defense and emitted an explicit warning. (2) **Missed-Detection Rate (MDR)**: The fraction of corrupted queries that bypassed the filtering mechanism. (3) **False-Alarm Rate (FAR)**: The fraction of nominal (clean) queries erroneously flagged as anomalous by the defense mechanism.

Results and Analysis. Table 5 summarizes the system’s robustness. The agent achieved a high overall DFR of 93.0% on corrupted data, isolating extreme physical outliers with a 96.0% interception rate. Crucially, on the nominal baseline, the False-Alarm Rate (FAR) was contained at 2.0%. This demonstrates that the IQR bounds defined in Algorithm 1 are sufficiently calibrated to preserve regular operational queries while forming a rigid safety boundary.

Table 5. Robustness metrics under extreme noise injection, including false-alarm and missed- detection rates.

Test Group	Total Queries	DFR (%)	MDR (%)	FAR (%)
Nominal baseline (clean)	100	N/A	N/A	2.0
Temporal dropout (20% loss)	50	90.0	10.0	N/A
Physical outliers (extreme values)	50	96.0	4.0	N/A
Overall Corrupted Set	100	93.0	7.0	N/A

3.7. Experiment 6: Quantitative Analysis of the Interference Hypothesis

To systematically test the Interference Hypothesis, we conducted a controlled ablation isolating the impact of RAG context size on internal reasoning coherence.

Experimental Design. We sampled 50 logic-heavy tasks from the Core Navigation domain. The MARTIAN model was evaluated under varying retrieval sizes: $K \in \{0, 3, 6, 10\}$. For each run, the intermediate reasoning tokens generated before the final action output were extracted.

Evaluation Metrics. Interference was quantified using two parameters: (1) **Reasoning Perplexity (PPL)**: The average perplexity of the generated reasoning tokens, functioning as an indicator of internal generative coherence. The PPL was computed using the evaluated model’s own next-token prediction logits, strictly over the intermediate Chain-of-Thought sequence prior to the final action output. (2) **Logical Breakage Rate**: The frequency with which the model retrieved the correct procedural rule text but generated an action command contradicting that specific rule owing to logic fragmentation.

Results and Analysis. Table 6 quantifies the degradation of coherence on procedural tasks as retrieval context expands. Under the $K = 10$ configuration, the logical breakage rate increased to 22.0%, accompanied by a significant rise in perplexity. This supports the assertion that unstructured external text introduces semantic interference in strictly logical procedures already internalized via supervised fine-tuning.

Table 6. Impact of RAG context size on reasoning coherence in strictly logical tasks.

Configuration	Accuracy (%)	Mean CoT PPL	Logical Breakage Rate (%)
$K = 0$ (SFT only)	78.5	4.12	N/A
$K = 3$ (Standard RAG)	77.3	5.34	8.5
$K = 6$ (Medium RAG)	72.9	6.71	13.8
$K = 10$ (Overloaded RAG)	62.1	8.76	22.0

4. Discussion and Conclusions

4.1. The Synergy of SFT and RAG in Maritime AI

This study provides empirical evidence resolving a critical architectural debate in domain-specific LLMs: the balance between internalized parameters and external retrieval. Our findings suggest a functional dichotomy. **Supervised Fine-Tuning (SFT)** acts as the agent’s reflexive memory, effectively embedding the procedural heuristics required for dynamic tasks such as collision avoidance (COLREGs), where latency and conditional logic are paramount. Conversely, **Retrieval-Augmented Generation (RAG)** functions as the declarative memory, indispensable for precision-critical inquiries (e.g., safety thresholds) where the cost of hallucination is prohibitive. However, the observed performance regression in reasoning-heavy models (e.g., DeepSeek-R1) when RAG is enabled highlights a non-trivial interference effect. This implies that future maritime agents should employ an adaptive routing mechanism, activating retrieval only for factual queries while relying on internalized parameters for logical deduction.

4.2. The Interference Hypothesis: Interaction Between RAG and Reasoning Models

A pivotal observation of this study, as shown in Table 2, is the performance variation of the reasoning-optimized DeepSeek-R1 model when augmented with RAG (decreasing from 69.26% to 66.01%). This trend contrasts with general-purpose models (e.g., DeepSeek-V3), which exhibited accuracy gains of over 14% with RAG enabled. We tentatively hypothesize that this may stem from a misalignment between *internalized reasoning chains* and *external context integration*. Reasoning-heavy models such as DeepSeek-R1 rely significantly on self-generated CoT tokens to structure their deductive logic. It is possible that the injection of retrieved text chunks—if not perfectly aligned with the model’s internal reasoning format—could introduce semantic fragmentation that interrupts the coherence of the generated thought process, effectively acting as semantic interference in specific logical tasks. This finding suggests a potential architectural nuance for Maritime AI: **selective retrieval**. Rather than a default static continuous retrieval approach, future systems might benefit from a routing layer that distinguishes between *logical reasoning tasks* (e.g., COLREG interpretation, where internalized parameters often suffice) and *knowledge retrieval tasks* (e.g., querying technical specifications, where external grounding is mandatory). This distinction offers a plausible explanation for why SFT-only MARTIAN outperformed RAG-enhanced variants in dynamic maneuvering scenarios.

4.3. Scalability and Maintenance of the RAG Knowledge Base

To ensure the long-term scalability and maintainability of the RAG component, the knowledge base employs a modular chunking strategy with metadata tagging (e.g., regulation version and effective date). Updates can be performed incrementally by re-indexing only changed documents (e.g., new IMO resolutions or flag state amendments), minimizing recomputation costs. Version control follows semantic versioning principles, with change logs and automated diff-based re-embedding triggered upon regulatory updates. Challenges such as semantic drift in evolving regulations are addressed through periodic human-in-the-loop validation and hybrid retrieval (keyword + dense).

4.4. Regulatory Compliance and Certification Pathway

Since the system is proposed for MASS, it is essential to acknowledge the regulatory context, though a comprehensive legal analysis is beyond the scope of this work. In brief, MARTIAN is positioned at IMO MASS Degree 2–3 (remotely controlled/automated decision support). Certification would follow interim guidelines from classification societies (e.g., Lloyd’s Register 2024 MASS framework, ClassNK AiP processes) and the forthcoming IMO MASS Code (non-mandatory 2025, mandatory ~2032). We note three key open challenges that warrant dedicated future study: (i) allocation of liability for erroneous AI recommendations (potentially shared between operator, system developer, and flag state), (ii) compliance pathways under adapted COLREG/STCW/SOLAS provisions, and (iii) cybersecurity requirements per IMO MSC-FAL.1/Circ.3. Our CoT-based explainability—visible reasoning traces grounded in retrieved regulations—is designed to facilitate, though not guarantee, the transparency requirements anticipated by these frameworks. Full certification pathways remain under active discussion by IMO LEG/MSC committees.

4.5. Implications for On-Board Deployment

From an engineering perspective, this work validates the feasibility of deploying high-performance cognitive agents without reliance on satellite-linked cloud infrastructure. By quantizing a 14B-parameter model to run within 6.5 GB of memory, we demonstrate that standard vessel workstations can support expert-level reasoning. This challenges the prevailing assumption that intelligent decision support requires massive, proprietary

foundation models, thereby offering a cost-effective pathway for retrofitting autonomous capabilities onto existing merchant fleets.

4.6. Limitations and Future Directions

Despite these advancements, our evaluation of the function calling interface revealed a cognitive synthesis gap: while the MARTIAN engine excels at discrete tool execution, it struggles to autonomously orchestrate multi-step sequences to resolve ambiguous, high-level goals. This suggests that the current paradigm of instruction tuning is sufficient for *execution* but insufficient for strategic *planning*.

Future work will address this limitation through two primary avenues:

1. **Reinforcement Learning for Planning:** Implementing Reinforcement Learning on Chain-of-Thought traces (RL-CoT) to specifically reward valid multi-step reasoning paths and penalize synthesis failures.
2. **Real-World Validation:** Moving beyond simulation to deploy the MARTIAN engine on a physical MASS test bed, validating its robustness against the noisy, unpredictable sensor data characteristic of open-water environments.

In conclusion, MARTIAN represents an initial engineering step towards deployable maritime decision support, demonstrating that a strategically specialized mid-size model can narrow the gap between raw perception and structured professional reasoning. While further validation in open-water environments is required before operational certification, the results provide a concrete reference architecture for edge-deployed cognitive agents on resource-constrained vessels.

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Abbreviations

The following abbreviations are used in this manuscript:

LLM	Large Language Model
MASS	Maritime Autonomous Surface Ships
RAG	Retrieval-Augmented Generation
CoT	Chain-of-Thought
SFT	Supervised Fine-Tuning
COLREGs	International Regulations for Preventing Collisions at Sea
AIS	Automatic Identification System
STCW	Standards of Training, Certification and Watchkeeping
DP	Dynamic Positioning
IMO	International Maritime Organization

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